

Coconut presentation outline

Joey David, M2 IASD

1. Motivation and background
 - 1.1. Reminder of what instruct-finetuned models are, of problem-solving challenges with LLMs
 - 1.2. Introduce/explanation: briefly what CoT is, how it works, how it's implemented
 - 1.3. The limitations of CoT Coconut aims to solve: break free from token space, reason in latent space insane. Gain computational efficiency, comparison with float $\leftrightarrow$ int
2. Implementation
 - 2.1. Core idea: don't translate back to token space during reasoning, feed the hidden state of the last pass directly to the current pass
 - 2.2 Modes of operation: instead of `<cot> </cot>`, `<bot> <eot>`
 - 2.3 training: gradually replace CoT steps with continuous thought, backprop on latent thoughts, mask loss on latent segments
3. Inference and emergent behaviors
 - 3.1 Inference: feed hidden states recursively, stop via classifier or length
 - 3.2 Emergent tree search: coconut encodes multiple alternative paths, behaves like BFS instead of cot's DFS
4. Empirical validation and results, implementation
 - 4.1 show performance on datasets, interpret
 - 4.2 explain cot, no-cot, iCot, Pause token
 - 4.3 ablations
 - 4.4 efficiency gains
5. Conclusion and future paths